CERVICAL CYTOLOGY-HISTOLOGY CORRELATION QUALITY ASSURANCE REPORT

Per ASC Clinical Practice Committee Birdsong Guideline 2017
CAP Accreditation Checklist Requirement CYP.06600

Total Cases	Concordant	Major Discordant	HSIL PV+
130	31	34	42.3%
cases analyzed	23.8%	26.1%	Target: >60%

CAP ALERT: Major discordance rate of 26.1% exceeds the CAP benchmark of less than 10%. Corrective action documentation required per CYP.06600.

PV+ ALERT: HSIL positive predictive value of 42.3% falls below the CAP target of 60%. Review HSIL overcall rate.

MAJOR FALSE NEGATIVES: 14 major undercall cases identified. All require mandatory slide review per Birdsong Section 8.

Executive Summary

A total of 130 cases were analyzed.

Section 1 - Concordance Summary

Classification of all cytology-histology pairs per ASC Birdsong concordance definitions.

Concordance_Class	Count	Percent
minor_discordant	65	50.0
major_discordant	34	26.15
concordant	31	23.85

Agreement Within One Grade (Birdsong Section 5)

Percentage of pairs with exact agreement or minor disagreement only. Per Birdsong, this metric is of educational interest but not statistically meaningful due to verification bias.

Metric	Count	Percentage
Agreement within one grade (concordant + minor)	**96**	73.8%
Exact agreement only	**31**	23.9%
Major discordance (outside one grade)	**34**	26.1%

Section 2 - Discrepancy Bucket Summary

CAP-style discrepancy classification per Birdsong Figure 1 and Figure 3. Undercall = cytology less severe than histology. Overcall = cytology more severe than histology.

Bucket	Count	Percent
MinVar	7	5.38
MajUnd	14	10.77
MinUnd	43	33.08
Agree	31	23.85
MinOver	15	11.54
MajOver	20	15.38

Section 3 - HSIL-Focused Metrics (Birdsong Section 7)

The five statistical calculations required by Birdsong Section 7 and the ASC Clinical Practice Committee. These are the primary performance metrics for cytopathology quality assurance.

Metric	Value
HSIL Pap tests with histologic HSIL (CIN2/CIN3)	42.31
HSIL Pap tests with minor discrepancies	23.08
HSIL Pap tests with major false-positive discrepancies	34.62
Number of major false-negative / cytology undercall discrepancies	14.0
PV+ for HSIL Pap tests (%)	42.31

PV+ Interpretation (Birdsong Section 7a)

LOW - Below CAP target of 60%. Review HSIL overcall rate.

Extended PV+ - High-Grade Cytology Group (HSIL + ASC-H + AGC-NEO)

Per Birdsong Figure 1, HSIL, ASC-H, and AGC-NEO cytology diagnoses are grouped together for concordance assessment. The extended PV+ reflects combined performance of the entire high-grade cytology group.

Metric	**Value**
Total high-grade Pap tests (HSIL + ASC-H + AGC-NEO)	**61**
High-grade Pap tests with HSIL+ histology	**31**
Extended PV+	**50.8%**

Section 4 - Intergrade Follow-Up Rates (Birdsong Section 8)

Per Birdsong Section 8: careful review of rates of HSIL histology following cytologic interpretations of ASC-US, ASC-H, and LSIL might reveal opportunities for improvement. Elevated rates may indicate systematic cytologic undercalling.

Cytology	Total Cases	HSIL+ Follow-up	HSIL+ Rate (%)
ASC-US	11	9	81.8
ASC-H	11	3	27.3
LSIL	17	11	64.7
AGC	14	10	71.4
AGC-NEO	9	5	55.6
AGC-EMC	4	4	100.0

Clinical interpretation: ASC-US to HSIL+ rates above 15-20% or LSIL to HSIL+ rates above 20-25% may indicate undercalling patterns requiring targeted educational slide review.

Section 5 - Key Quality Signals

Summary of key performance indicators with CAP benchmark comparisons. Status reflects comparison against published CAP and ASC thresholds.

Quality Signal	Value	CAP Threshold	Status
Major discordance rate	26.1%	< 10%	EXCEEDS
Minor discordance rate	50.0%	No threshold	-
Undercall proportion	43.9%	< 15%	EXCEEDS
Overcall proportion	26.9%	No threshold	-
HSIL PV+ (positive predictive value)	42.3%	> 60%	LOW
Extended PV+ (HSIL + ASC-H + AGC-NEO)	50.8%	> 60%	LOW
Agreement within one grade	73.8%	Educational interest	-
Major false negative count	14	Minimize	REVIEW

Section 6 - Visualizations

All figures generated by the CHC-QA Pipeline per ASC Birdsong Guideline 2017.

Figure 1 - Concordance Distribution

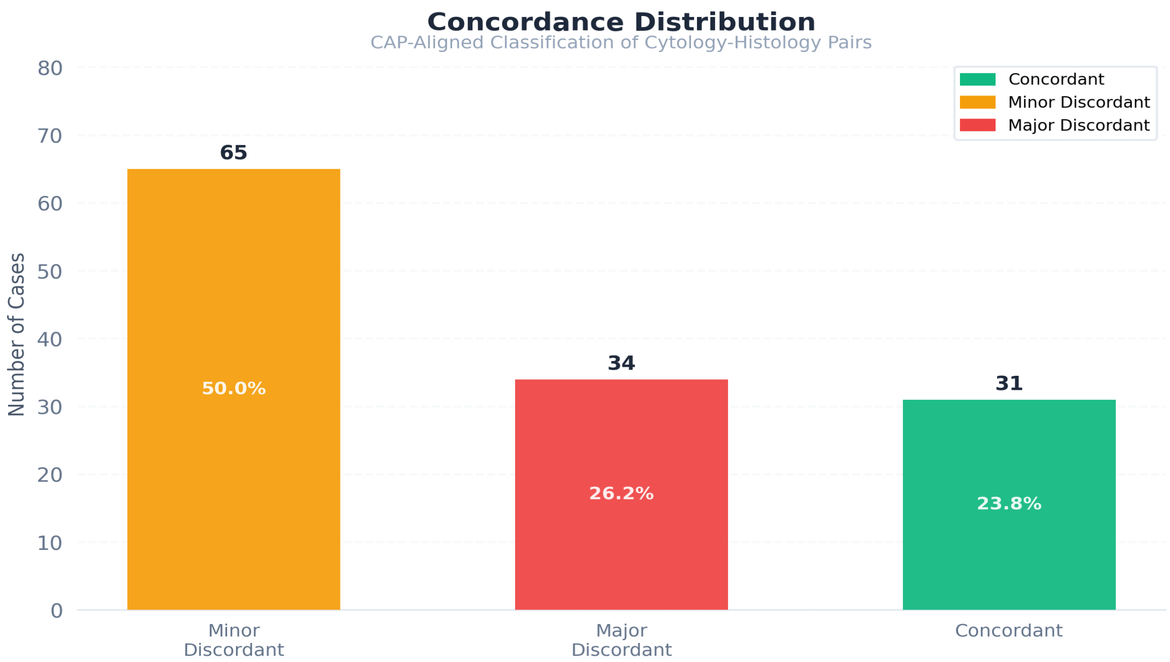


Figure 2 - Cytology-Histology Correlation Buckets

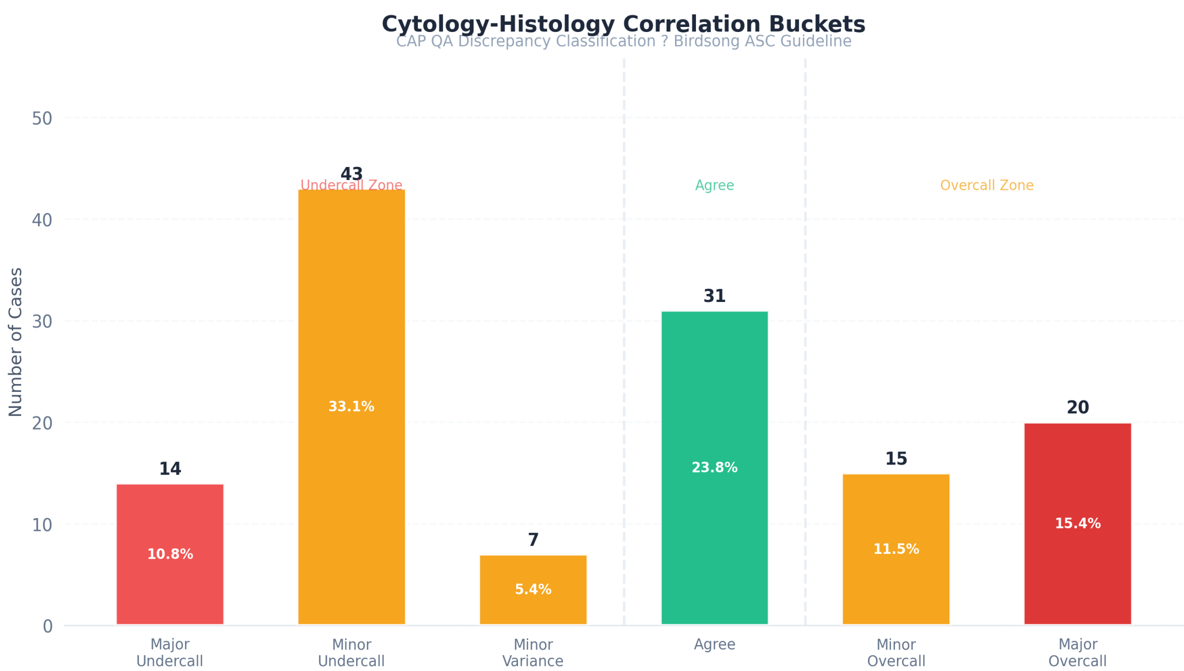


Figure 3 - HSIL Correlation Metrics

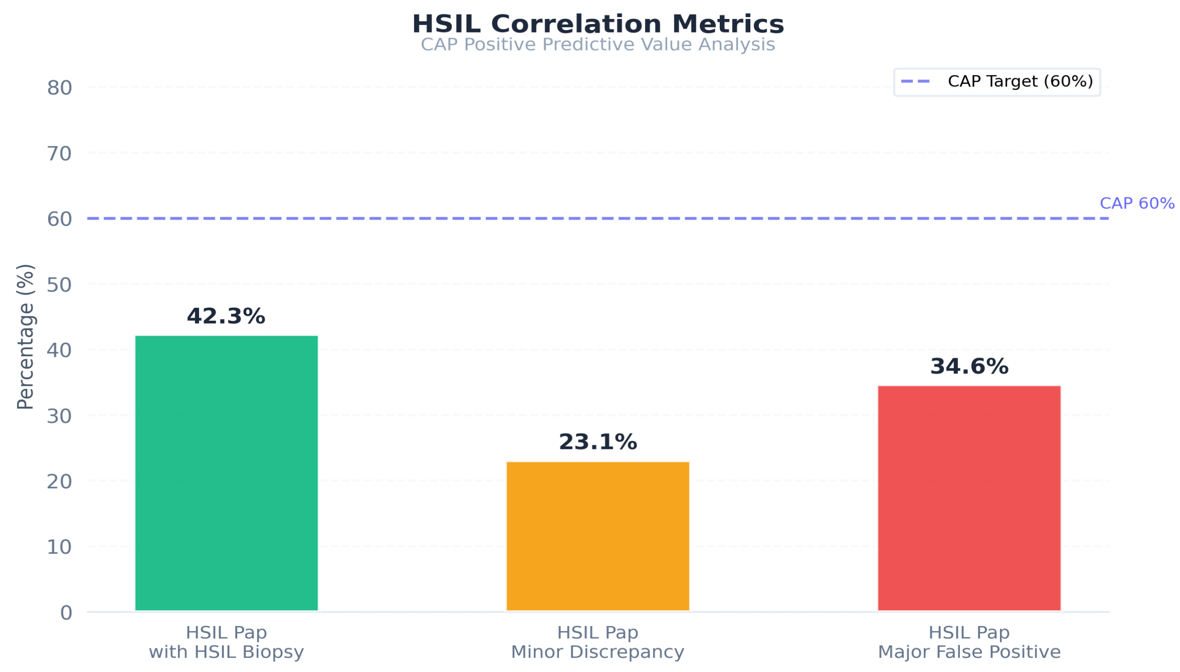


Figure 4 - Cytology vs Histology Confusion Matrix

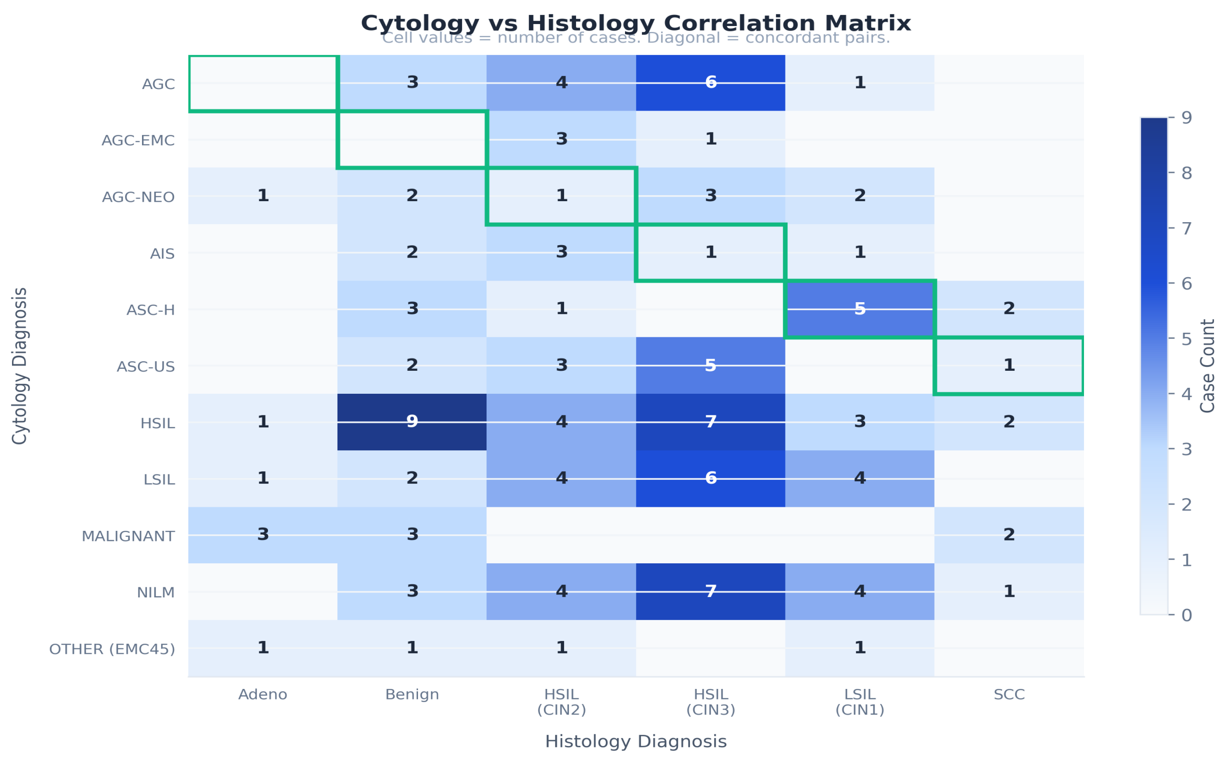
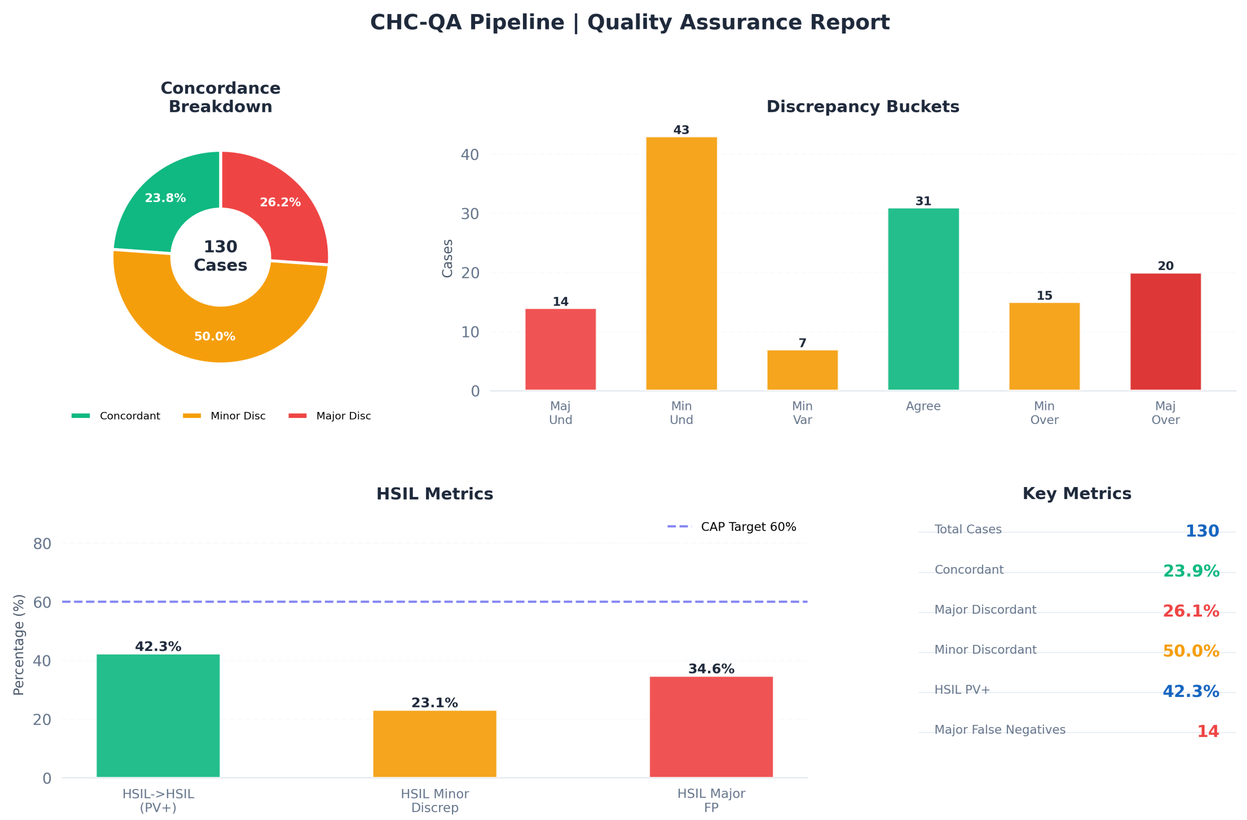


Figure 5 - Summary Dashboard



Section 7 - AI Clinical Commentary

The following commentary was generated by a locally-hosted AI language model based on the QA metrics above. This commentary should be reviewed by a qualified cytopathologist before use in official documentation.

Paragraph 1 - PATTERN ANALYSIS: The laboratory's performance demonstrates a significant discordance issue, with a major discordance rate of 26.2% substantially exceeding the CAP benchmark of less than 10%. Analysis reveals a pattern of overcalling, as evidenced by an overcall proportion of 26.9% and a notable number of major overcalls (20) and minor overcalls (15). Furthermore, the undercall proportion of 43.9% combined with a substantial number of minor undercalls (43) contributes to the overall discordance. These figures strongly suggest a systematic issue impacting the accuracy of diagnoses, particularly within higher-grade lesions.

Paragraph 2 - ROOT CAUSE HYPOTHESIS: Based on Birdsong Section 8, interpretive error appears to be the most probable contributing factor, supported by the high major discordance rate and the 50.0% minor discordance rate. The undercall proportion of 43.9% may also indicate interpretive bias, where pathologists are hesitant to confirm HSIL diagnoses. Screening error, potentially related to inadequate sampling, is suggested by the undercall proportion, and sampling error could be a contributing factor given the 14 major undercalls. Finally, differences of opinion may exist, but the magnitude of the discordance suggests a primary interpretive issue.

Paragraph 3 - BENCHMARK INTERPRETATION: The laboratory's performance falls significantly short of CAP benchmarks. The major discordance rate of 26.2% is substantially above the acceptable threshold, indicating a serious deficiency in diagnostic accuracy. The low HSIL PV+ rate of 42.3% is below the CAP target of 60%, and the Australian Royal College standard of greater than 65%, raising concerns about the detection of clinically significant abnormalities. Consequently, immediate corrective action is required to address this widespread discordance and ensure diagnostic reliability.

Paragraph 4 - INTERGRADE ANALYSIS: The intergrade follow-up rates reveal concerning trends. The elevated HSIL+ follow-up rates for ASC-US (81.8%), LSIL (64.7%), and AGC (71.4%) suggest a potential overdiagnosis of HSIL within these categories. Notably, the rates for ASC-H (27.3%) and AGC-NEO (55.6%) are below expected ranges, potentially indicating underdiagnosis of HSIL in these groups. These findings highlight the need for a more nuanced approach to grading and a thorough review of the diagnostic criteria applied to each case.

Paragraph 5 - CORRECTIVE ACTION RECOMMENDATIONS: To address these deficiencies, we recommend a prioritized action plan. First, implement CAP CYP.06600, requiring a second pathologist review of all HSIL cases and a comprehensive review of all cases with discordant interpretations (Birdsong Section 8, Step 3). Second, conduct a structured

Section 8 - Limitations

This report is generated by an automated deterministic cytology-histology correlation pipeline. The following limitations apply and should be considered when interpreting results.

1. **Verification bias:** CHC datasets are subject to verification bias due to selective histologic follow-up. Results should not be interpreted as measures of screening sensitivity or specificity.
2. **Clinical context:** Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.
3. **LLM normalization:** Free-text diagnosis normalization was performed by a locally-hosted large language model. Normalized terms were validated against a controlled vocabulary. Residual normalization errors may affect a small number of cases.
4. **Pairing logic:** Cases were paired by patient identifier and date window. Complex scenarios including multiple biopsies, multiple Pap tests, and delayed follow-up may result in suboptimal pairing in a minority of cases.
5. **Root cause classification:** This pipeline classifies discordance type and direction but does not perform root cause analysis (sampling error, screening error, interpretive error). Manual review of major discordant cases is required per Birdsong Section 8.

Reference: Birdsong GG, Walker JW. Gynecologic Cytology-Histology Correlation Guideline. ASC Bulletin. 2017;LIV(2):VIII-XIII. | CAP Checklist Requirement CYP.06600 | Generated by CHC-QA Pipeline (github.com/solomoneluanu/chc-qa-pipeline)